# TRACE AND NORM ARE TRANSITIVE IN TOWERS OF FINITE EXTENSIONS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: The transitivity half of the trace/norm formalism, companion to the minimal-polynomial formulas (b10 card L48387).

*Editorial note.* The label lemma-trace-and-norm-tower names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Chapter lemma-trace-and-norm-linear (source lemma, worklist row 20): for a finite-dimensional L-vector space V and an L-linear map phi, Trace_K(phi) = Trace_(L/K)(Trace_L(phi)) and det_K(phi) = Norm_(L/K)(det_L(phi))
- Chapter definitions: the trace and norm of an element alpha in M over L are the trace and determinant of the L-linear multiplication map m_alpha on M

*Source reference.* Original reference 0BII: sources/stacks-fields/pinned-original-source.tex, source lines 2661–2661. Prerequisite pointer invoked at 2724; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *Let $M/L/K$ be a tower of finite extensions of fields. Then*

$$\mathit{Trace}_{M/K} = \mathit{Trace}_{L/K} \circ \mathit{Trace}_{M/L} \quad \text{and} \quad \mathit{Norm}_{M/K} = \mathit{Norm}_{L/K} \circ \mathit{Norm}_{M/L}$$

*Proof.* Think of $M$ as a vector space over $L$ and apply Lemma 0BII. □

*Date*: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.